Polar Measurements on a Small Turntable: How Well Can the Rotation Center Be Corrected?

A free-field simulation study of a horizontal three-way monitor

Technical report — simulation code and data:
https://github.com/Arnwald/turntable-rotation-center-study

Abstract

A polar measurement should rotate the loudspeaker about a reference point on its baffle. The turntables available to small manufacturers and hobbyists are usually too small for that: the cabinet ends up centered on the turntable, the rotation axis lies well behind the baffle and, for an asymmetric layout, off to the side. The microphone is then farther away and in another direction than intended. A purely geometric correction — each driver treated as a point source with its own directivity, measurements re-propagated to the intended distance and interpolated to the intended direction — can rebuild the intended polar data. We test it in free field on a simulated horizontal three-way monitor (8-inch woofer, 3-inch dome midrange, 1.1-inch dome tweeter, $440 \times 290 \times 290$ mm closed box) whose cabinet diffraction follows the Distributed Edge Dipole model, for a horizontal orbit from $-180°$ to $180°$ in $10°$ steps at 1 m and 2 m. Without diffraction the correction is essentially exact (below 0.1 dB RMS). With diffraction, about 0.24 dB RMS remains on the summed loudspeaker at 1 m (2.45 dB without correction), concentrated behind the cabinet at mid and high frequencies. The Predicted In-Room Response computed from the corrected data stays within 0.14 dB (1 m) and 0.07 dB (2 m) of the rigorous one, instead of 1.02 dB and 0.48 dB uncorrected. The lateral offset of the microphone is essential: omitting it or entering it with the wrong sign does more harm than not correcting at all.

Contents

1 Introduction **2**

2 The simulated loudspeaker **3**
2.1 Cabinet and layout . . . 3
2.2 Driver responses from the datasheets . . . 3
2.3 Radiators . . . 4
2.4 Cabinet diffraction: the Distributed Edge Dipole model . . . 5
2.5 An illustrative three-way crossover . . . 6

3 Measurement geometry **6**
3.1 Turntable kinematics . . . 6
3.2 What the raw measurement gets wrong . . . 7

4 The correction method **9**
4.1 Inputs and geometry . . . 9
4.2 Point-source re-propagation . . . 9
4.3 Angular interpolation . . . 10
4.4 Implementation . . . 10
4.5 Assumptions . . . 10

5 Simulation protocol **10**

6 Results **11**
6.1 Uncorrected: what the centered setup does to the data 11
6.2 Corrected, direct sound only: the method itself 13
6.3 Corrected, with diffraction: the open question 13
6.4 Angular resolution 13
6.5 Spatial averages and Predicted In-Room Response 16
6.6 Wrong parameters and the laterally shifted microphone 16

7 Discussion **19**

8 Conclusion **20**

A Additional figures **20**

B Reproducibility **21**

1 Introduction

The off-axis behavior of a loudspeaker is measured by turning it in front of a fixed microphone. Standards such as CTA-2034-A place the center of rotation on the baffle, at the reference point (usually the tweeter axis), so that the microphone moves on a circle of constant radius around it. With a large turntable, or a rig that holds the cabinet with its baffle over the axis, that is easy. The turntables most small manufacturers and hobbyists can afford (a few hundred millimeters across) cannot carry a cabinet that overhangs by 30 to 45 cm: the cabinet is centered on the turntable, and the rotation axis passes through the middle of the box (Figure 4).

Two things then go wrong. Off axis, the microphone is no longer at the intended distance from the drivers: farther away behind the cabinet, so the off-axis responses are too low and too late. And the direction in which each driver is seen is no longer the nominal angle. When the drivers are not on the cabinet center line — a horizontal monitor with its woofer beside the midrange-tweeter column, as here — the microphone has to be moved sideways to face the reference point at 0°, and the errors become asymmetric.

A correction can be built from geometry alone. Around each driver, a point source with an arbitrary directivity is assumed; every measurement is moved to the intended distance (a delay and a gain) and the response in the intended direction is interpolated between the two measurements that bracket it. The method needs four numbers per plane: where the rotation axis was relative to the baffle (depth and lateral offset), where each driver is relative to the reference axis, and the microphone distance.

This report checks how well that works. The setting is deliberately idealized: free field (no room, no gating), a loudspeaker simulated from its drivers' datasheets, and a diffraction model for the cabinet edges, which is the open question — the edges radiate from places that are not the driver, so a point-source correction cannot be exact for them. We compare, at 1 m and 2 m, the rigorous measurement (axis on the baffle), the raw measurement with the cabinet centered on the turntable, and the same measurement after correction, driver by driver, for a summed three-way loudspeaker and for the CTA-2034-A spatial averages, Predicted In-Room Response included. We also look at what happens when the correction parameters are wrong, which is the practical risk.

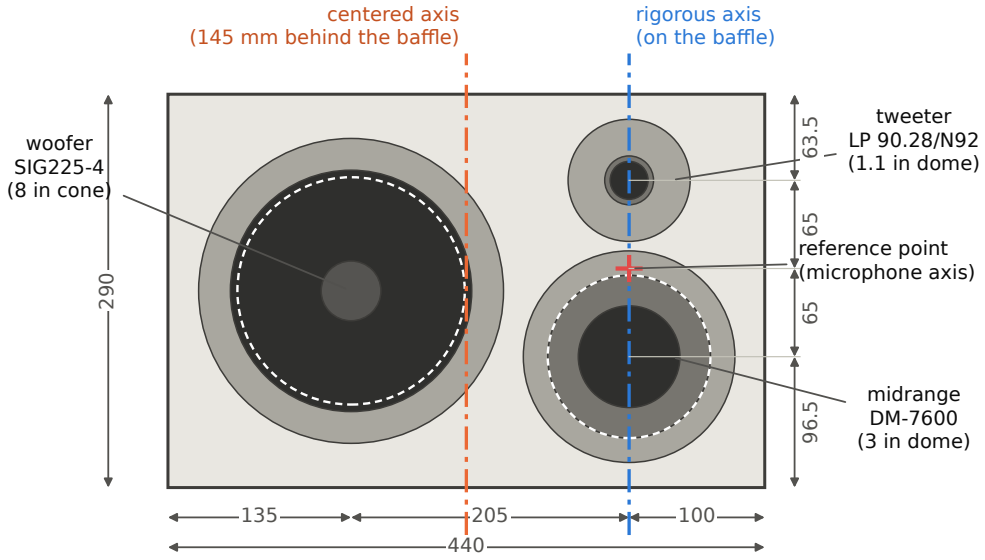


Figure 1: Front view of the simulated monitor (dimensions in mm). Dashed white circles: radiators used in the model (piston of the woofer's S_d, equivalent radiating cap of the midrange). Blue: rotation axis of the rigorous setup, on the baffle through midrange and tweeter. Orange: rotation axis when the cabinet is centered on the turntable, 145 mm behind the baffle.

Table 1: Driver positions in the speaker frame and radiator models.

	x (mm)	y (mm)	radiator
Woofer, Dayton Audio SIG225-4	−85	0	flat piston, $a = \sqrt{S_d/\pi} = 83.8\,\text{mm}$
Midrange, HiVi DM-7600	120	−48.5	equivalent cap, $a = 60\,\text{mm}$, $h = 17.5\,\text{mm}$
Tweeter, SICA LP 90.28/N92	120	81.5	spherical cap, $a = 14\,\text{mm}$, $h = 7.5\,\text{mm}$
Reference point **R**	120	16.5	—

2 The simulated loudspeaker

2.1 Cabinet and layout

The loudspeaker is a horizontal three-way monitor in a closed box of $W \times H \times D = 440 \times 290 \times 290\,\text{mm}$ (24 L net), Figure 1. The woofer is on the left; the midrange and the tweeter are stacked on the right, tweeter on top. All positions are expressed in the *speaker frame*: origin at the center of the front baffle, x to the right as seen from the listening position, y up, z normal to the baffle toward the listener. The cabinet occupies $-D \le z \le 0$ and every driver sits on the baffle plane $z = 0$ (Table 1). The *reference point* **R** is on the baffle, on the vertical line through the midrange and tweeter centers, halfway between them; the *reference axis* is the baffle normal through **R**. The microphone is placed on that axis, at the height of **R**.

2.2 Driver responses from the datasheets

Each driver is described by its on-axis half-space response $H_i(f)$: complex pressure at $r_{\text{ref}} = 1\,\text{m}$ for 2.83 V, in an infinite baffle. The SPL curves $L_i(f)$ of the three datasheets are vector graphics; they were extracted point by point from the PDF files rather than traced by eye (the CSV files are in the repository). Each curve is taken as the half-space response, interpolated linearly in $\log f$ and extended outside its published range with fixed slopes s_{lo} and s_{hi} (dB per octave):

$$L(f) = L(f_{\min}) + s_{\text{lo}} \log_2 \frac{f}{f_{\min}} \quad (f < f_{\min}), \qquad L(f) = L(f_{\max}) + s_{\text{hi}} \log_2 \frac{f}{f_{\max}} \quad (f > f_{\max}), \tag{1}$$

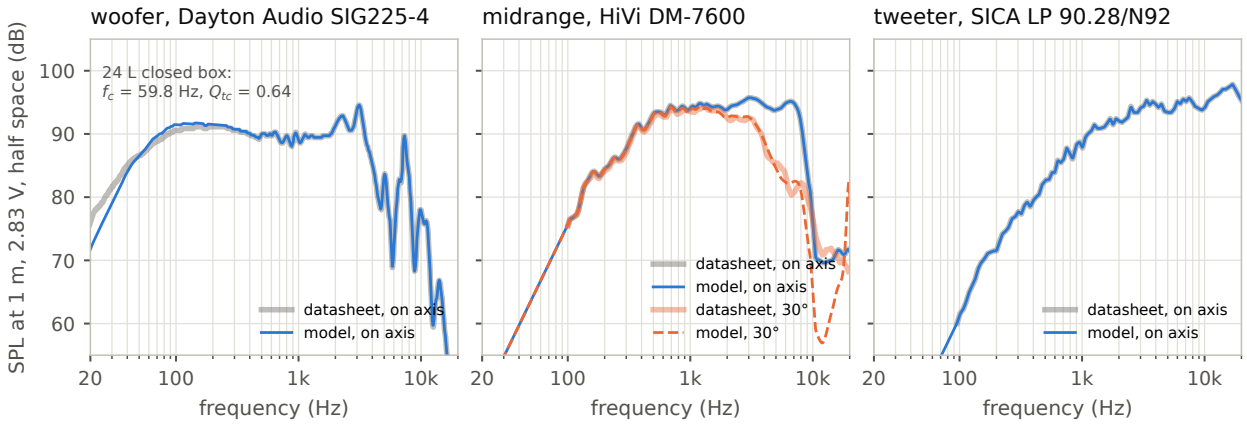


Figure 2: Driver models against the datasheet curves (half space, 1 m, 2.83 V). The woofer is shown in its 24 L closed box. For the midrange, the datasheet also gives the 30° response, which the equivalent radiating cap reproduces to within 0.8 dB RMS between 1.5 and 8 kHz.

with $s_{\mathrm{lo}} = +12$ for all three drivers and $s_{\mathrm{hi}} = -24$ (woofer) or -12 (domes). The magnitude $|H_i| = p_{\mathrm{ref}}\, 10^{L_i/20}$ ($p_{\mathrm{ref}} = 20\,\mu\mathrm{Pa}$) is given a minimum phase with the folded real cepstrum: on an N-point DFT grid,

$$c[n] = \mathrm{IDFT}\{\ln|H[k]|\}, \qquad \hat{c}[n] = \begin{cases} c[n] & n = 0,\ n = N/2 \\ 2c[n] & 0 < n < N/2 \\ 0 & N/2 < n < N \end{cases}, \qquad H_{\min}[k] = \exp\left(\mathrm{DFT}\{\hat{c}\}[k]\right), \tag{2}$$

evaluated on a grid eight times finer than the simulation grid to avoid cepstral aliasing.

Woofer in its closed box. The SIG225-4 datasheet curve is close to an infinite-baffle response at low frequency (its low end follows $f_s = 40\,\mathrm{Hz}$, $Q_{ts} = 0.43$). It is moved into the 24 L box with the ratio of the two second-order high-pass functions,

$$H_{\mathrm{woofer}}(f) = H_{\mathrm{ds}}(f)\,\frac{\mathcal{H}(f; f_c, Q_{tc})}{\mathcal{H}(f; f_s, Q_{ts})}, \qquad \mathcal{H}(f; f_0, Q) = \frac{s^2}{s^2 + s/Q + 1},\ s = \mathrm{j}\frac{f}{f_0}, \tag{3}$$

$$\alpha = \frac{V_{as}}{V_b} = \frac{29.7}{24}, \qquad f_c = f_s\sqrt{1+\alpha} = 59.8\ \mathrm{Hz}, \qquad Q_{tc} = Q_{ts}\sqrt{1+\alpha} = 0.64 \tag{4}$$

(lossless box). Both high-pass functions are minimum phase, so the product stays minimum phase. As a check, the Thiele–Small parameters give a mass-controlled sensitivity

$$p = \frac{\rho_0\, S_d\, Bl\, V}{2\pi r\, R_e\, M_{ms}} = 0.75\ \mathrm{Pa} \quad (\rho_0 = 1.204\ \mathrm{kg/m^3}), \tag{5}$$

that is 91.5 dB at 1 m for the datasheet's 91.4 dB.

Band limit. All responses are multiplied by a raised-cosine taper from 20.5 to 23.5 kHz, which keeps the impulse responses band-limited at the 48 kHz sample rate.

Figure 2 compares models and datasheets.

2.3 Radiators

The directivities are the classical far-field ones [4], normalized to 1 on the driver's own axis (θ is the angle from that axis, $k = 2\pi f/c$, $c = 343\,\mathrm{m/s}$).

Cone: rigid flat piston of radius $a = \sqrt{S_d/\pi}$ in an infinite baffle,

$$D_{\mathrm{p}}(\theta, k) = \frac{2J_1(ka\sin\theta)}{ka\sin\theta}. \tag{6}$$

Dome: rigid convex spherical cap of base radius a and height h, moving axially. Its far field is the Rayleigh integral over the projected disc, each ring of radius r being advanced by the cap height $z(r)$:

$$A(\theta, k) = \frac{2}{a^2}\int_0^a J_0(kr\sin\theta)\, e^{\mathrm{j}kz(r)\cos\theta}\, r\,\mathrm{d}r, \qquad z(r) = \sqrt{R_s^2 - r^2} - (R_s - h), \quad R_s = \frac{a^2+h^2}{2h}, \tag{7}$$

$$D_{\mathrm{d}}(\theta, k) = \frac{A(\theta, k)}{A(0, k)}. \tag{8}$$

D_{d} is complex: the protruding center moves the apparent source off axis. With $h = 0$ it is the piston. The integral is evaluated with 96-point Gauss–Legendre quadrature.

The midrange's equivalent cap. The DM-7600 sits in a horn-shaped aluminum faceplate. A bare 75 mm dome is far wider than the datasheet's 30° curve (3 dB down at 5 kHz where the datasheet shows 8 dB). Fitting (a, h) of Eq. (7) to the datasheet difference $L_{30^\circ} - L_{0^\circ}$ between 1.5 and 8 kHz gives $a = 60$ mm, $h = 17.5$ mm (0.83 dB RMS): a 120 mm aperture, the diameter of the faceplate's mouth (the baffle cutout is 120.5 mm). That equivalent cap is used for the midrange. The tweeter, with no off-axis data, uses its own geometry: a 28 mm silk dome, assumed 7.5 mm high.

2.4 Cabinet diffraction: the Distributed Edge Dipole model

The edges of the front baffle are modelled with the Distributed Edge Dipole (DED) approach of Urban *et al.* [1], a refinement of the edge-source picture of Vanderkooy [3]: a line of dipoles along the edges, driven by the pressure the driver produces there. In the form used here, extended to a source of finite size, driver i produces at the microphone position $\mathbf{M}$

$$p_i(\mathbf{M}, f) = H_i(f)\,\left[P_{\mathrm{dir},i}(\mathbf{M}, f) + P_{\mathrm{edge},i}(\mathbf{M}, f)\right], \tag{9}$$

$$P_{\mathrm{dir},i} = D_i(\theta_i, k)\,\frac{1+\cos\theta_i}{2}\,\frac{r_{\mathrm{ref}}}{r_i}\,e^{-\mathrm{j}kr_i}, \qquad r_i = |\mathbf{M} - \mathbf{S}_i|, \quad \cos\theta_i = \frac{\mathbf{n}\cdot(\mathbf{M}-\mathbf{S}_i)}{r_i}, \tag{10}$$

$$P_{\mathrm{edge},i} = \sum_e Q_{i,e}(f)\,F_e(\mathbf{M})\,\frac{r_{\mathrm{ref}}}{r_{2,e}}\,e^{-\mathrm{j}kr_{2,e}}, \qquad r_{2,e} = |\mathbf{M} - \mathbf{E}_e|, \quad F_e(\mathbf{M}) = -\frac{\mathbf{n}\cdot(\mathbf{M}-\mathbf{E}_e)}{r_{2,e}}, \tag{11}$$

where $\mathbf{S}_i$ is the driver center, $\mathbf{n} = \mathbf{e}_z$ the baffle normal, D_i the radiator directivity, and $\mathbf{E}_e$, Δl_e the midpoints and lengths of the elements along the four edges of the baffle (2 mm elements, 730 in all). The factor $(1+\cos\theta)/2$ is the obliquity of the direct field: 1 on axis, $\frac{1}{2}$ in the baffle plane, 0 behind; F_e is the edge dipole, zero in the baffle plane and changing sign behind it. The strength of edge element e is

$$Q_{i,e}(f) = C_i\,\Delta l_e\,\frac{G_i(\rho_{i,e}, f)}{A_i(0,k)}, \qquad G_i(\rho, f) = \frac{1}{S_i}\iint_{S_i}\frac{1+\cos\theta_s}{2}\,\frac{e^{-\mathrm{j}kr_1}}{r_1}\,\mathrm{d}S, \tag{12}$$

where the integral runs over the radiating surface (the projected disc, each point raised by the cap height $z(r)$ for a dome), r_1 is the distance from the surface point to the edge point, $\cos\theta_s = -z(r)/r_1$ the obliquity toward it, $\rho_{i,e} = |\mathbf{E}_e - \mathbf{S}_i|$, and $A_i(0,k)$ (Eq. 7; 1 for a piston) refers the edge drive to the on-axis level that H_i describes. The constant

$$C_i = \frac{1}{2\sum_e \Delta l_e\, G_i(\rho_{i,e}, 0)} \tag{13}$$

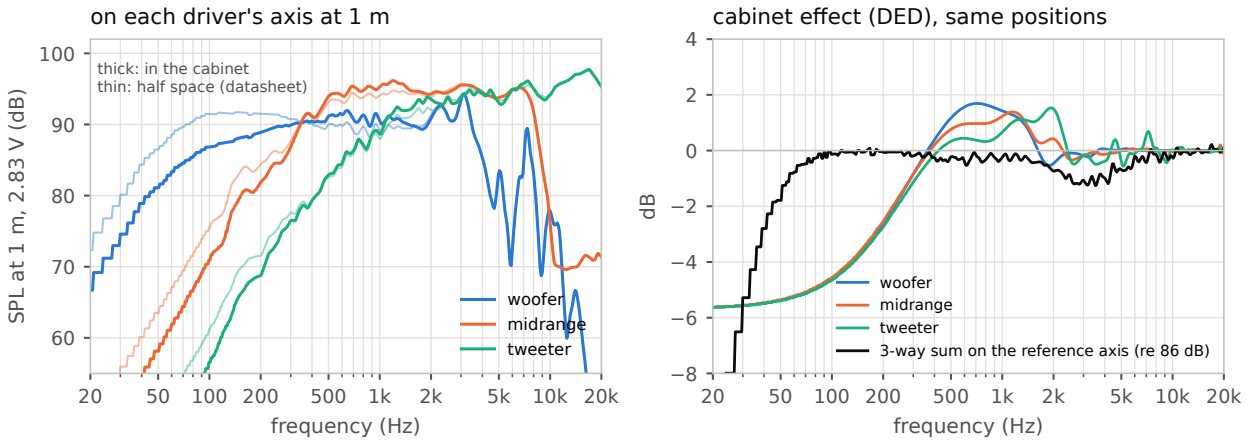


Figure 3: Left: each driver on its own axis at 1 m, in the cabinet (thick) and in half space (thin). Right: their ratio, the cabinet effect according to the DED model (baffle step and edge ripple), and the illustrative three-way sum on the reference axis.

makes $P_{\text{dir}} + P_{\text{edge}} = \frac{1}{2}$ on axis in the far field at low frequency: the cabinet radiates 6 dB below the half-space response at low frequency (full-space radiation) and reaches it at high frequency — the baffle step.

All paths keep their true length: the edge waves arrive $(r_1 + r_2 - r)/c$ after the direct sound, *at the actual microphone position*. That finite-distance geometry is exactly what a point-source correction cannot reproduce, and the reason for using the DED model here. As in the DED formulation, only the edges of the face that carries the drivers are modelled; behind the baffle the field is the sum of the edge dipoles plus the vanishing direct term.

Numerics. $G_i(\rho, f)\, e^{\mathrm{j}k\rho}$ varies slowly with ρ and f once the common delay ρ/c is removed. It is tabulated on a 1 mm grid of ρ and a 50 Hz grid of f, with a Gauss–Legendre rule in r and a midpoint rule in azimuth over half a turn (the integrand is even), sized for the phase spread $2k_{\max}a$ across the surface ($n_r = \lceil 0.7 \cdot 2k_{\max}a + 16 \rceil$, $n_\varphi = \lceil 0.5 \cdot 2k_{\max}a + 16 \rceil$); it is then interpolated linearly in ρ and with cubic splines in f. On each driver's axis at 30 m, $P_{\text{dir}} + P_{\text{edge}}$ is at -5.97 dB at 20 Hz for all three drivers, and within 0.3 dB of 0 dB at 10 kHz.

2.5 An illustrative three-way crossover

To look at the summed loudspeaker, each driver is equalized to an acoustic target on the reference axis at 1 m (rigorous setup):

$$T_{\text{woofer}} = \mathrm{LR}_4^{\mathrm{LP}}(450)\,\mathrm{B}_2^{\mathrm{HP}}(40), \quad T_{\text{mid}} = \mathrm{LR}_4^{\mathrm{HP}}(450)\,\mathrm{LR}_4^{\mathrm{LP}}(3000), \quad T_{\text{tweeter}} = \mathrm{LR}_4^{\mathrm{HP}}(3000), \quad (14)$$

with $\mathrm{B}_2^{\mathrm{LP}}(f_0) = 1/(s^2 + \sqrt{2}s + 1)$, $\mathrm{B}_2^{\mathrm{HP}} = s^2/(s^2 + \sqrt{2}s + 1)$, $s = \mathrm{j}f/f_0$, and $\mathrm{LR}_4 = \mathrm{B}_2^2$. The filter of driver i is the minimum-phase filter (Eq. 2) with magnitude $L\,|T_i|/S_i$, where S_i is the 1/6-octave smoothed on-axis magnitude of the driver and L corresponds to 86 dB SPL. The summed response is $p_{\text{sys}} = \sum_i X_i\, p_i$. The same filters are used in every case: only the measurements change. The design is not optimized (Figure 3, right) and does not need to be.

3 Measurement geometry

3.1 Turntable kinematics

The turntable axis is vertical, at $\mathbf{A} = (a_x, \cdot, a_z)$ in the speaker frame. The microphone is fixed in the room; at 0° it is on the reference axis, at distance d from the baffle: $\mathbf{M}_0 = (x_R,\, y_R,\, d)$. Turning

(a) rigorous: the axis runs along the baffle, through midrange and tweeter

(b) small turntable: the axis runs through the middle of the cabinet

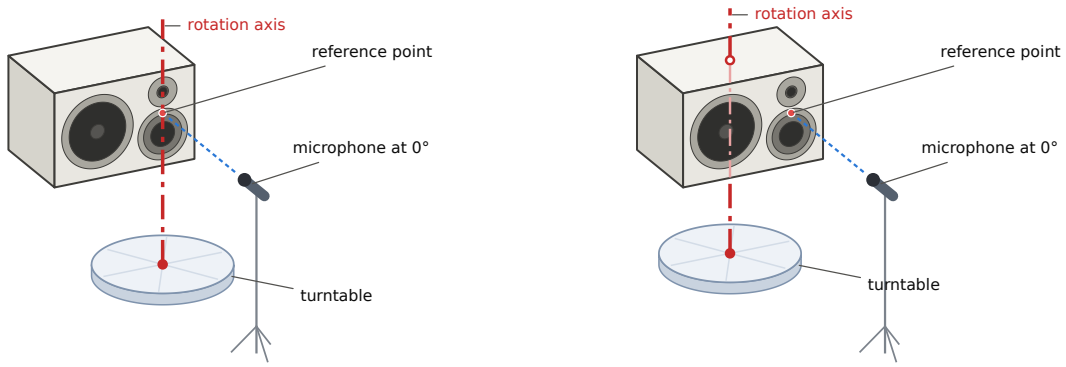


Figure 4: The two setups, with the loudspeaker raised above the disc that stands for the turntable. Dash-dotted red: the rotation axis — (a) along the baffle, through the midrange and the tweeter; (b) through the middle of the cabinet (lighter where it is hidden inside the box). The microphone faces the reference point (red dot) at 0° in both cases.

Table 2: Range of the geometric errors of the raw measurement (centered setup).

driver	d	level (dB)	delay (ms)	direction (°)
tweeter / midrange	1 m	−2.54 to +0.33	−0.11 to +0.99	−3.4 to +15.4
woofer	1 m	−2.81 to +0.17	−0.06 to +0.94	−5.2 to +14.4
tweeter / midrange	2 m	−1.35 to +0.17	−0.12 to +0.98	−1.8 to +8.2
woofer	2 m	−1.41 to +0.13	−0.09 to +0.95	−2.3 to +7.9

the loudspeaker by θ moves the microphone, seen from the loudspeaker, by $-\theta$ about the axis. With positive angles bringing the microphone toward $+x$ (the midrange/tweeter side):

$$\begin{aligned} x(\theta) &= a_x + (x_R - a_x)\cos\theta + (d - a_z)\sin\theta, \\ z(\theta) &= a_z - (x_R - a_x)\sin\theta + (d - a_z)\cos\theta, \end{aligned} \qquad y(\theta) = y_R. \tag{15}$$

- **Rigorous setup**: **A** on the reference point, $(a_x, a_z) = (x_R, 0)$. The microphone stays on the circle of radius d around **R**: $(x, z) = (x_R + d\sin\theta,\ d\cos\theta)$.
- **Centered setup**: the cabinet is centered on the turntable, $(a_x, a_z) = (0, -D/2)$, i.e. 145 mm behind the baffle and 120 mm to the left of the reference axis. The microphone has been moved 120 mm sideways so that it faces the reference point at 0° (Figures 4 and 5).

3.2 What the raw measurement gets wrong

Seen from a driver at **S**, the microphone of the centered setup is at a distance $\rho_k = |\mathbf{M}(\theta_k) - \mathbf{S}|$ and in a direction φ_k that differ from the intended ones (Figure 6). Figure 7 and Table 2 give the consequences in the horizontal plane: the raw off-axis responses are up to 2.5–2.8 dB too low and almost 1 ms late behind the cabinet at 1 m, and the direction is wrong by up to 15°. At 2 m the level and direction errors halve; the delay does not (it is set by the axis offset, not by the distance).

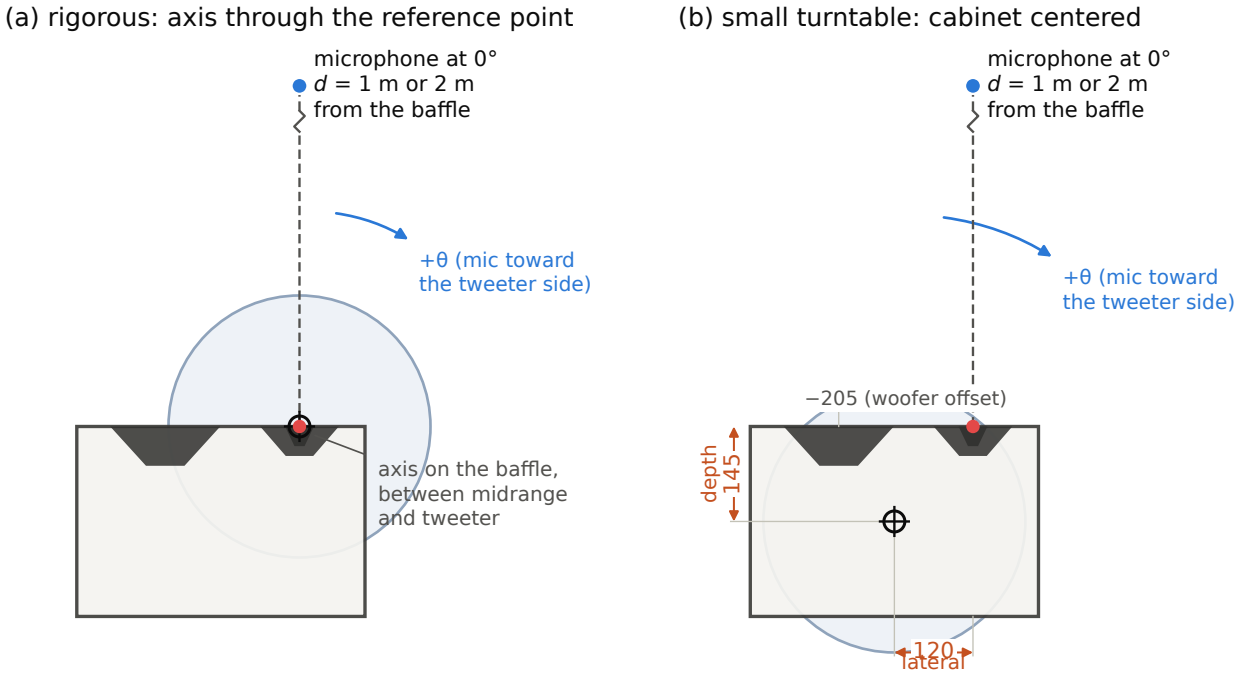


Figure 5: Top views to scale (turntable 400 mm across; microphone distance not to scale). In (b) the correction needs the depth (rotation axis to baffle, 145 mm), the lateral offset (rotation axis to reference axis, +120 mm, positive toward the side the microphone faces at +90°), each driver's offset from the reference axis (−205 mm for the woofer, 0 for midrange and tweeter) and the microphone distance.

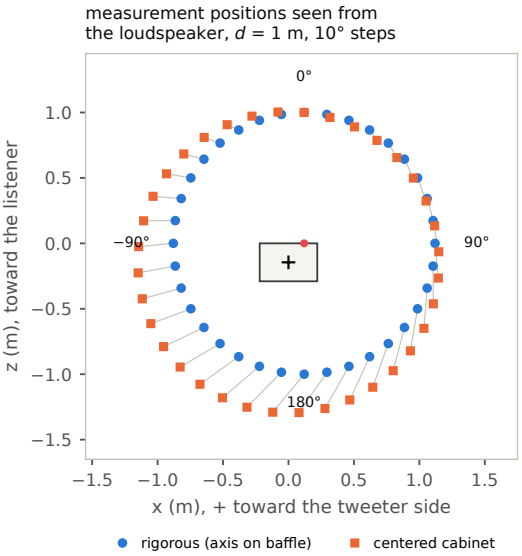


Figure 6: Microphone positions relative to the loudspeaker, to scale, at 1 m. Blue: rigorous setup, a circle of 1 m around the reference point. Orange: centered setup, a circle of 1.15 m around the middle of the cabinet. Gray segments join the positions with the same nominal angle.

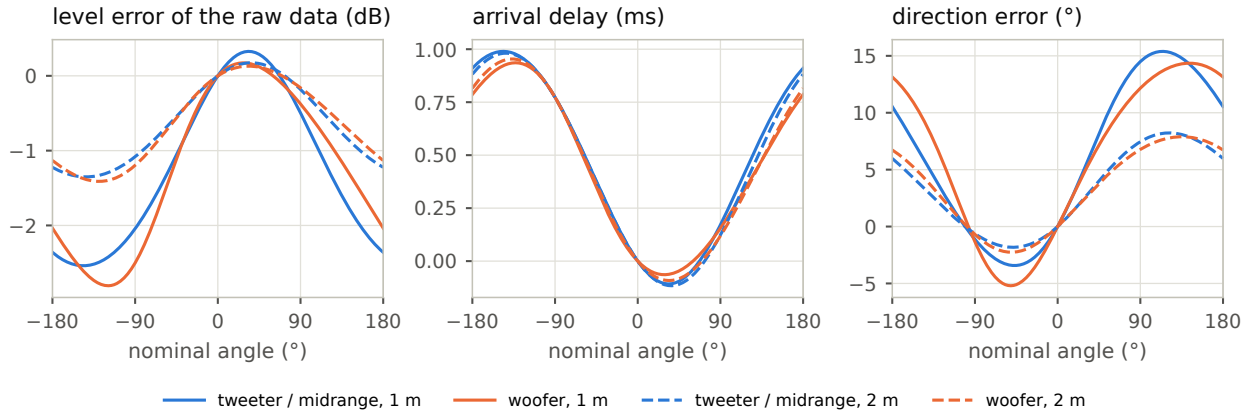


Figure 7: Errors of the raw measurement in the centered setup, seen from the drivers: level ($20\log_{10}$ of the distance ratio), extra arrival delay and direction error, against the nominal angle.

4 The correction method

The correction rebuilds, for each nominal angle, the response that would have been measured at the ideal position. It works plane by plane and driver by driver.

4.1 Inputs and geometry

In the plane of rotation, take a frame with the rotation axis at the origin, y toward the front and x toward the *+ side* (the side the microphone faces at +90°). The user provides

- the depth δ: from the rotation axis to the baffle (positive when the axis is behind it);
- the lateral offset ℓ: from the rotation axis to the reference axis, positive toward the + side;
- the driver offset o: from the reference axis to the driver, positive toward the + side;
- the microphone distance r_0 from the baffle at 0°.

Then the reference point is $\mathbf{R} = (\ell, \delta)$, the driver $\mathbf{S} = \mathbf{R} + (o, 0)$, and

$$\text{measured position:} \quad \mathbf{M}_k = \mathcal{R}(\theta_k)\,(\ell,\ \delta + r_0), \tag{16}$$

$$\text{ideal position:} \quad \mathbf{M}^*(\theta) = \mathbf{R} + r_0(\sin\theta,\ \cos\theta), \tag{17}$$

with the rotation $\mathcal{R}(\theta)(x, y) = (x\cos\theta + y\sin\theta,\ -x\sin\theta + y\cos\theta)$ (positive angles toward the + side). The 0° measurement is on the reference axis, ℓ away from the rotation axis: the lateral shift of the microphone is part of the model. Seen from the driver, every position has a direction and a distance,

$$\varphi = \operatorname{atan2}(M_x - S_x,\ M_y - S_y) \ \text{ (unwrapped to within } \pi \text{ of } \theta), \qquad \rho = |\mathbf{M} - \mathbf{S}|. \tag{18}$$

For the setup of Figure 5(b): $\delta = 145$ mm, $\ell = +120$ mm, $o = -205$ mm for the woofer and 0 for the midrange and tweeter, $r_0 = 1000$ or 2000 mm.

4.2 Point-source re-propagation

Around each driver the response is assumed to be that of a point source with a directivity,

$$h(\varphi, \rho, t) = \frac{g(\varphi,\ t - \rho/c)}{\rho}, \tag{19}$$

so a measurement taken at (φ_k, ρ_k) is moved to the target distance ρ^* by a delay and a gain:

$$h_{k\to *}(t) = \frac{\rho_k}{\rho^*}\, h_k\!\left(t - \frac{\rho^* - \rho_k}{c}\right). \tag{20}$$

4.3 Angular interpolation

The measurements, sorted by φ_k, serve as supports (for a full circle, copies at $\varphi_k \pm 2\pi$ close the loop). For the target direction φ^* the two bracketing supports a, b are interpolated linearly *after* both have been moved to ρ^*, so that they are in time and do not comb-filter:

$$h^*(t) = (1-u)\, h_{a\to *}(t) + u\, h_{b\to *}(t), \qquad u = \frac{\varphi^* - \varphi_a}{\varphi_b - \varphi_a}. \tag{21}$$

Outside the measured range the nearest support is used. The 0° measurement is the reference and is never changed. A measurement copied by symmetry at $-\theta$ stands for the mirror image of the one at θ and is placed at $(-\varphi_k, \rho_k)$ of that one.

4.4 Implementation

The fractional delays are applied as linear phases. With $\tau_k = (\rho^* - \rho_k) f_s / c$ samples, $p = \lceil \max|\tau_k| \rceil + 1$ samples of margin and an FFT length $N_c = 2^{\lceil \log_2(L+2p+64) \rceil}$ (L the longest impulse response),

$$H^*[m] = \sum_{\text{supports}} g_k\, X_k[m]\, e^{-\mathrm{j}2\pi \tilde{m}(\tau_k + p)/N_c}, \qquad \tilde{m} = \begin{cases} m & m \le N_c/2 \\ m - N_c & m > N_c/2 \end{cases}, \tag{22}$$

with g_k the interpolation weight times ρ_k/ρ^* and X_k the DFT of the zero-padded measurement; the corrected impulse response is $\mathrm{Re}\,\mathrm{IDFT}\{H^*\}$ from sample p on.

4.5 Assumptions

The method assumes a point source on the baffle plane whose far-field directivity does not depend on distance, and ignores what is outside the plane of rotation. The cabinet edges violate the first assumption; the vertical offsets between drivers and microphone (here ±65 mm for the domes), the third.

5 Simulation protocol

Measurements. Horizontal orbit from −180° to 180° in 10° steps (36 positions; ±180° is one position), microphone at $d = 1$ m and $d = 2$ m from the baffle, at the reference height. For every position, Eq. (9) gives the complex spectrum of each driver on a 16 384-point grid at 48 kHz (2.93 Hz resolution); the impulse responses are its inverse real FFT (341 ms, far longer than any delay or decay).

Cases.

- **rigorous**: axis on the reference point — the reference;
- **uncorrected**: cabinet centered, microphone moved onto the reference axis;
- **corrected**: the same data after the correction with the true parameters;
- **direct sound only**: the three cases again with $P_{\text{edge}} = 0$, to separate the correction's own errors from those due to diffraction;
- **variants**: corrected with wrong parameters (lateral offset omitted or with the wrong sign, woofer offset omitted, depth or lateral offset off by ±10 mm, microphone distance off by +20 mm), and with a 5° orbit.

Table 3: Sound-power weights of the horizontal orbit (CTA-2034-A Appendix C), normalized over the 36 directions; $w(180^\circ - \theta) = w(\theta)$.

0°	0.0012	10°	0.0095	20°	0.0180	30°	0.0248	40°	0.0301
50°	0.0338	60°	0.0364	70°	0.0381	80°	0.0390	90°	0.0394

Metrics. Power spectra are smoothed to 1/24 octave,

$$P_X(\theta, f) = \left\langle |p_X(\theta, f')|^2 \right\rangle_{f' \in [f\,2^{-1/48},\, f\,2^{1/48}]}, \tag{23}$$

and compared with the rigorous case through the level error

$$\Delta L_X(\theta, f) = 10 \log_{10} \frac{P_X(\theta, f)}{P_{\text{rigorous}}(\theta, f)} \text{ dB}. \tag{24}$$

Statistics (RMS, 95th percentile, maximum of $|\Delta L|$) are taken over the points that matter: in the driver's band (woofer 50 Hz–1 kHz, midrange 300 Hz–6 kHz, tweeter 2–20 kHz, summed loudspeaker 50 Hz–20 kHz), and no more than 20 dB below the rigorous on-axis response at the same frequency (this leaves out the deep nulls of a beaming cone, where a fraction of a degree changes the level by tens of dB).

CTA-2034-A curves from the horizontal orbit. Only the horizontal orbit is measured, so the standard's curves [2] are estimated from it (all averages on squared pressures):

$$\text{LW}^2 = \tfrac{1}{7} \textstyle\sum_{\theta \in \{0, \pm 10, \pm 20, \pm 30\}} |p(\theta)|^2, \tag{25}$$

$$\text{ER}^2 = \tfrac{1}{3}(\text{FW}^2 + \text{SW}^2 + \text{RW}^2), \tag{26}$$

$$\text{SP}^2 = \frac{\sum_\theta w(\theta)\, |p(\theta)|^2}{\sum_\theta w(\theta)}, \tag{27}$$

$$\text{PIR}^2 = 0.12\,\text{LW}^2 + 0.44\,\text{ER}^2 + 0.44\,\text{SP}^2, \qquad \text{DI} = 10 \log_{10} \frac{\text{LW}^2}{\text{SP}^2}, \tag{28}$$

where the front-wall (FW), side-wall (SW) and rear-wall (RW) groups are themselves RMS averages over the angles 0, ±10, ±20, ±30° (FW), ±40 to ±80° (SW) and ±90, 180° (RW). The standard adds the ±10° vertical directions to the listening window and the floor and ceiling bounces (vertical orbit) to the early reflections; using the horizontal weights alone for the sound power amounts to assuming the vertical orbit equal to the horizontal one. The sound-power weights are those of the standard's Appendix C,

$$w(\theta) = \Omega(\theta + 5^\circ) - \Omega(\theta - 5^\circ), \qquad \Omega(\alpha) = 4C - 2\pi, \tag{29}$$

with $\Omega(\alpha)$ the solid angle of the spherical square bounded by the great circles at $\pm\alpha$ horizontally and vertically (C its corner angle), symmetric about 90° and doubled at 90° (Table 3). These estimates are the same for every case, so their differences measure the measurement geometry only.

6 Results

6.1 Uncorrected: what the centered setup does to the data

Figure 8 (center column) and Figure 9 show the uncorrected data against the rigorous ones. Off axis the responses are too low — by the distance ratio, up to 2.5 dB behind the cabinet at 1 m — and, where the response changes quickly with angle (beaming of the woofer and of the midrange's horn-loaded dome, diffraction ripple behind the cabinet), they show features of the wrong direction. Within the relevant points the summed loudspeaker is off by 2.45 dB RMS at 1 m and 1.24 dB at 2 m (Table 4).

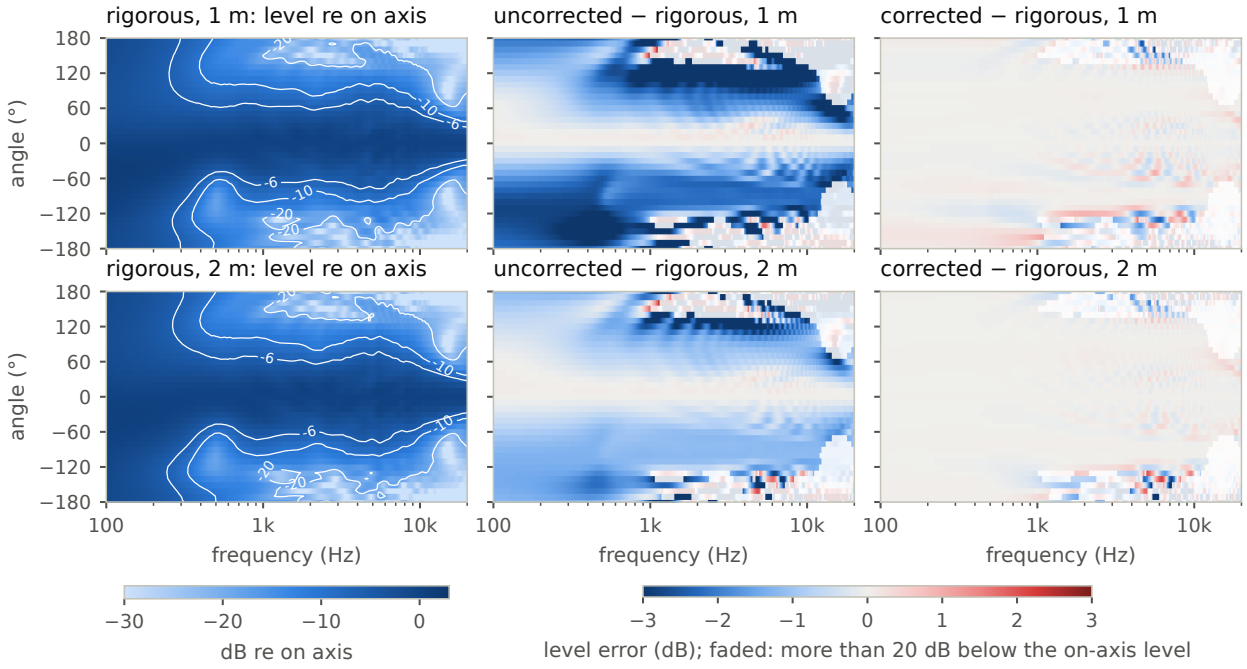


Figure 8: Summed loudspeaker. Left: rigorous horizontal polar map, normalized to the on-axis response (contours at −6, −10, −20 dB). Center and right: level error of the uncorrected and corrected data; points more than 20 dB below the on-axis level are faded.

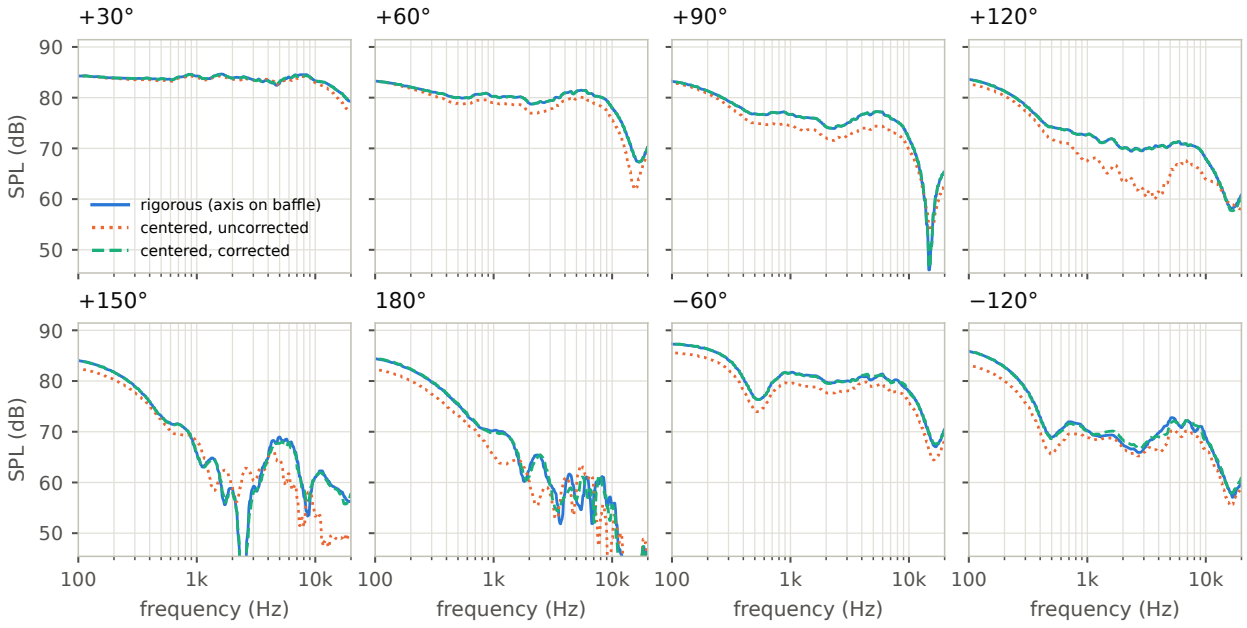


Figure 9: Summed loudspeaker at selected angles, 1 m, 1/12-octave smoothing.

Table 4: Level error $|\Delta L|$ against the rigorous measurement over the relevant points (dB): RMS, 95th percentile and maximum.

			uncorrected			corrected		
d	model	driver	RMS	p95	max	RMS	p95	max
1 m	DED	woofer	1.84	2.93	8.7	0.09	0.20	0.4
1 m	DED	midrange	2.39	4.27	20.1	0.21	0.38	2.1
1 m	DED	tweeter	3.30	6.79	22.8	0.40	0.96	3.2
1 m	DED	3-way sum	2.45	4.07	25.6	0.24	0.42	3.1
1 m	direct only	woofer	2.95	7.10	9.1	0.03	0.07	0.1
1 m	direct only	midrange	2.76	6.16	9.2	0.08	0.14	0.6
1 m	direct only	tweeter	2.72	5.93	9.2	0.05	0.09	0.3
1 m	direct only	3-way sum	2.77	6.30	9.3	0.03	0.07	0.3
2 m	DED	woofer	0.98	1.62	7.7	0.03	0.06	0.3
2 m	DED	midrange	1.19	2.09	9.2	0.14	0.26	1.2
2 m	DED	tweeter	1.71	3.57	12.0	0.41	0.74	5.4
2 m	DED	3-way sum	1.24	2.07	9.1	0.23	0.25	5.3
2 m	direct only	woofer	1.36	3.16	4.0	0.02	0.04	0.1
2 m	direct only	midrange	1.36	2.96	4.5	0.07	0.12	0.6
2 m	direct only	tweeter	1.34	2.88	4.5	0.04	0.07	0.4
2 m	direct only	3-way sum	1.33	3.16	4.3	0.03	0.05	0.4

6.2 Corrected, direct sound only: the method itself

Without edge diffraction every driver is exactly a point source with a directivity, as the correction assumes. What remains is the interpolation between measured directions and the vertical offsets that the planar model ignores: 0.02–0.08 dB RMS and 0.6 dB at worst (Table 4, "direct only"). Moving every driver to the microphone height changes the RMS figures by less than 0.01 dB: the out-of-plane offsets are negligible at 1 m and beyond. In this idealized world the correction is, for practical purposes, exact.

6.3 Corrected, with diffraction: the open question

With the cabinet edges, the residual grows: 0.40 dB RMS for the tweeter and 0.24 dB for the summed loudspeaker at 1 m, with local deviations up to 3.2 dB (Table 4, Figures 10 and 11). Figure 12 shows where: in the front half ($|\theta| \leq 90°$) no angle exceeds 0.3 dB RMS (0.2 dB for the summed loudspeaker); the residual is behind the cabinet ($|\theta| \geq 120°$), at mid and high frequencies, where the edge waves dominate the response — up to 1.3 dB RMS at one angle for the tweeter at 1 m, 3 dB at 2 m. The woofer, whose band is low, is hardly affected.

Two mechanisms are at work. First, the edges are not at the driver: the correction moves the edge waves by the driver's change of distance, not by their own, so their timing relative to the direct sound keeps the error of the measured position — visible in the late part of the impulse responses (Figure 13). This error shrinks with distance. Second, behind the cabinet the response is an interference pattern of many edge waves, with lobes about 10° wide at 5 kHz (λ/W) and narrower above; the correction interpolates between measured directions 10° apart, which cannot follow it. This one does not shrink with distance: at 2 m the summed loudspeaker keeps 0.23 dB RMS and its worst local deviation even grows to 5.3 dB, in a narrow region near −170° around 5–6 kHz.

6.4 Angular resolution

Halving the step to 5° addresses the second mechanism. At 2 m it brings the summed loudspeaker from 0.23 to 0.09 dB RMS and its worst deviation from 5.3 to 1.3 dB; the tweeter goes from 0.41 to

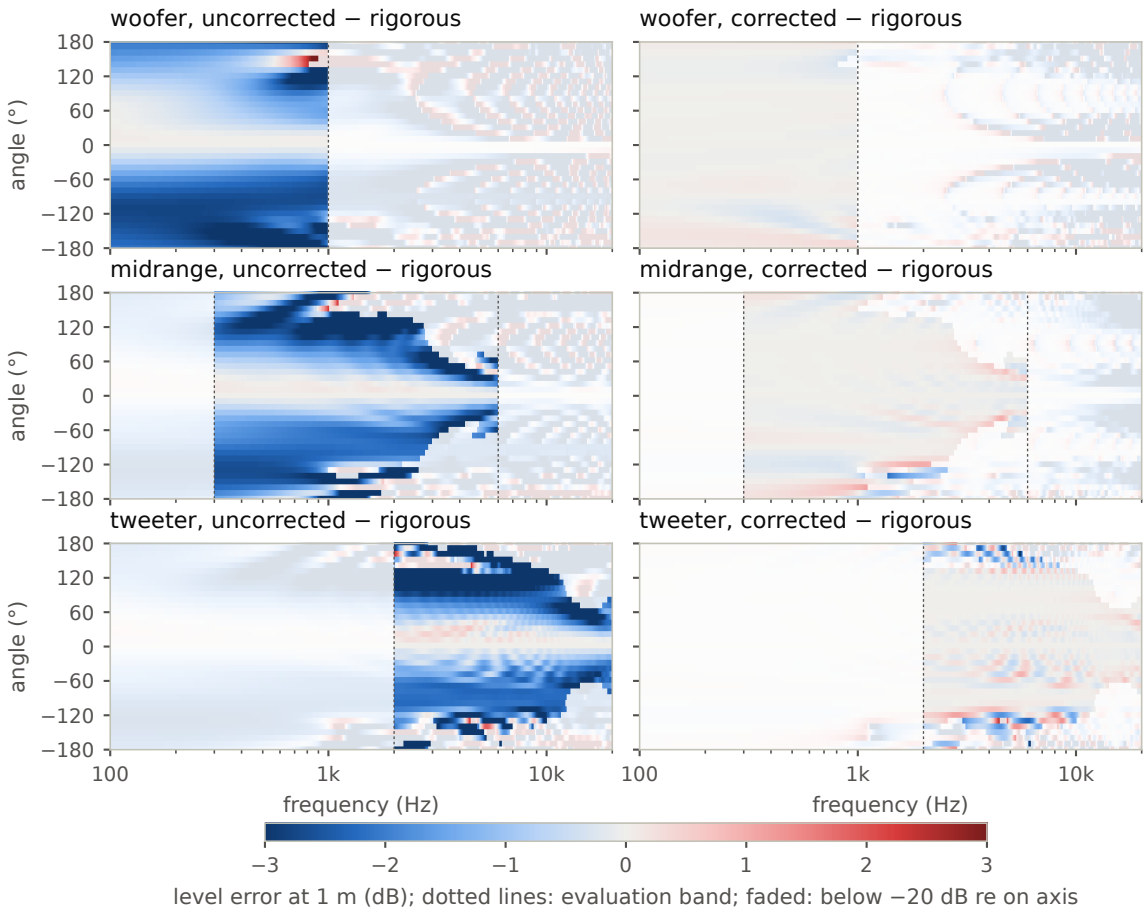


Figure 10: Level error of each driver at 1 m, uncorrected (left) and corrected (right). Dotted lines: evaluation band. Faded: more than 20 dB below the on-axis response.

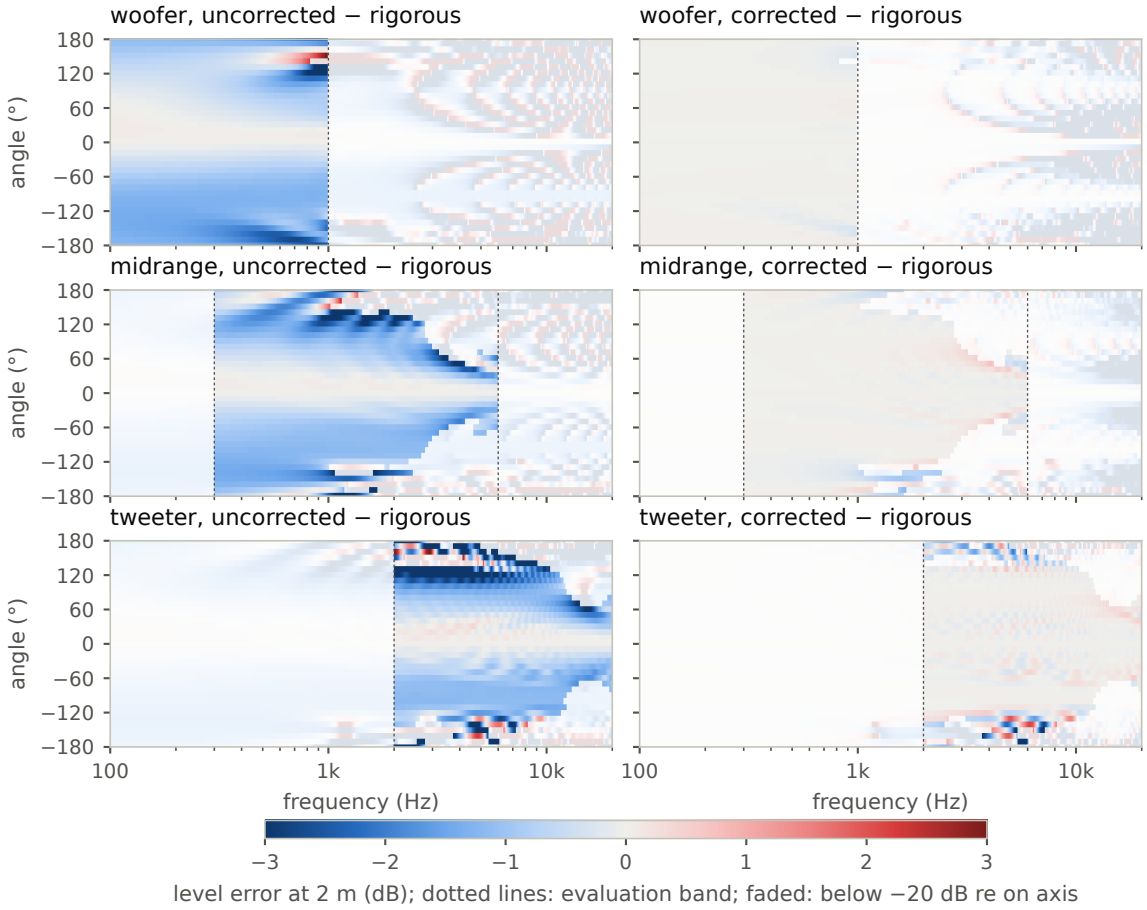


Figure 11: Same as Figure 10 at 2 m.

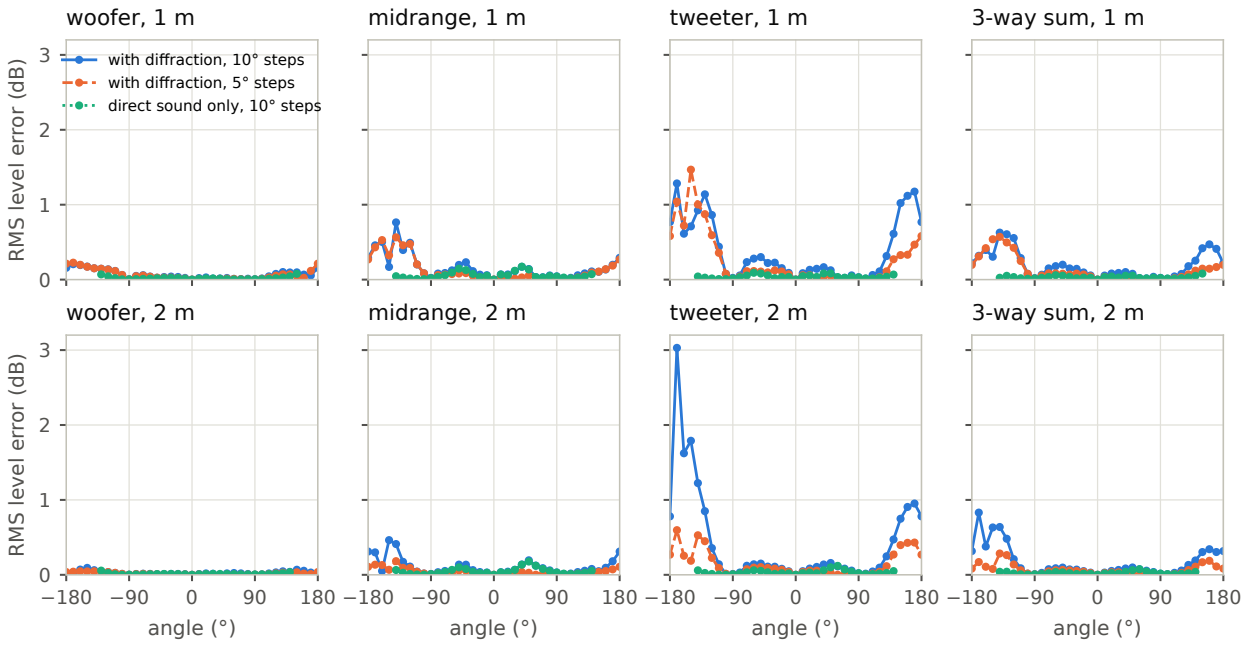


Figure 12: RMS level error of the corrected data at each angle (over the relevant frequencies), with the DED diffraction and 10° or 5° steps, and for the direct sound alone.

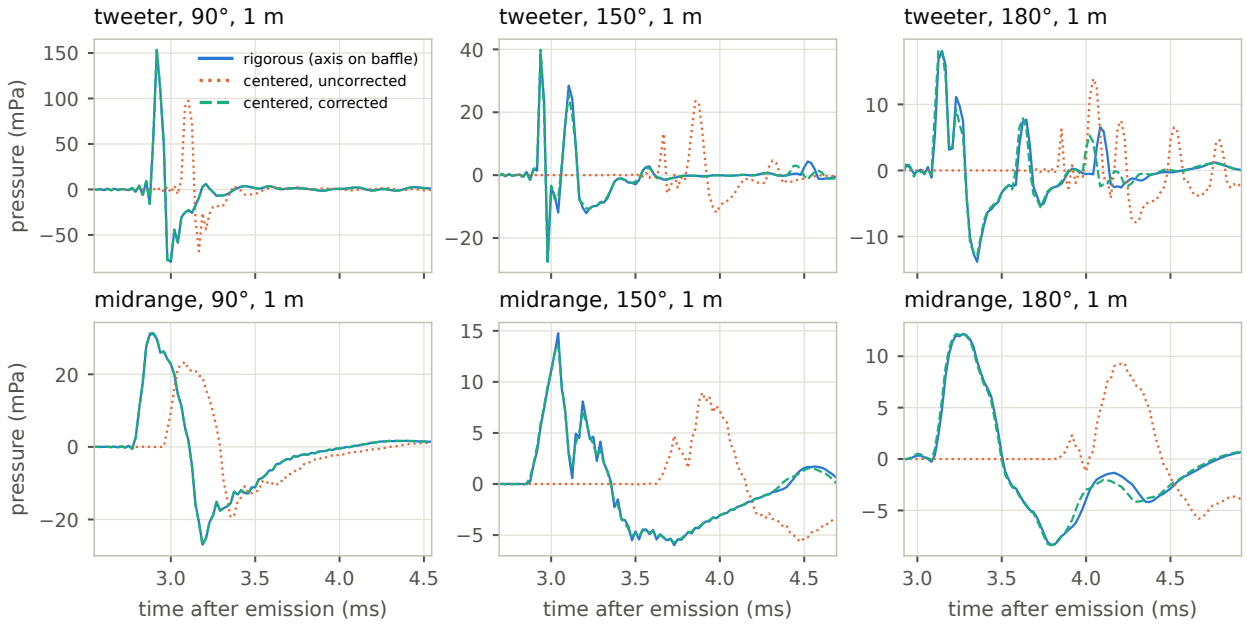


Figure 13: Impulse responses at 1 m (time from emission). The uncorrected responses arrive late and low; the corrected ones line up with the rigorous ones, except for the later edge-wave arrivals behind the cabinet.

Table 5: Largest deviation from the rigorous case, 100 Hz–20 kHz, of the horizontal-orbit CTA-2034-A curves of the summed loudspeaker (dB).

d	model	data	LW	ER	SP	PIR	DI
1 m	DED	uncorrected	0.51	0.95	1.39	1.02	1.19
1 m	DED	corrected	0.18	0.13	0.13	0.14	0.11
1 m	direct only	uncorrected	0.51	0.87	1.33	0.86	1.11
1 m	direct only	corrected	0.05	0.05	0.02	0.05	0.04
2 m	DED	uncorrected	0.26	0.44	0.66	0.48	0.58
2 m	DED	corrected	0.09	0.07	0.07	0.07	0.06
2 m	direct only	uncorrected	0.26	0.40	0.63	0.40	0.53
2 m	direct only	corrected	0.03	0.02	0.01	0.02	0.03

0.16 dB RMS. At 1 m the gain is smaller (0.24 to 0.19 dB RMS), because there the first mechanism, which no angular step can fix, takes a larger share.

6.5 Spatial averages and Predicted In-Room Response

The CTA-2034-A curves average many directions and the residual errors, concentrated behind the cabinet and changing sign from one frequency to the next, largely cancel (Figure 14, Table 5). Uncorrected, the centered setup lowers the sound power by up to 1.4 dB at 1 m (every off-axis microphone position is farther away), the PIR by up to 1.02 dB and shifts the directivity index by 1.2 dB; at 2 m these errors halve. Corrected, the PIR stays within 0.14 dB (1 m) and 0.07 dB (2 m) of the rigorous one, the sound power within 0.13 dB and the directivity index within 0.11 dB.

6.6 Wrong parameters and the laterally shifted microphone

The correction needs the geometry it is given (Figure 15, Table 6). Errors of ±10 mm on the depth or the lateral offset, or of 20 mm on the microphone distance, change the RMS errors by a few hundredths of a dB and the PIR by less than 0.1 dB: measuring with a tape is good enough.

The lateral offset itself is another matter. Because the microphone has been moved sideways to face the reference point, the model must know that the reference axis is $\ell = 120$ mm from

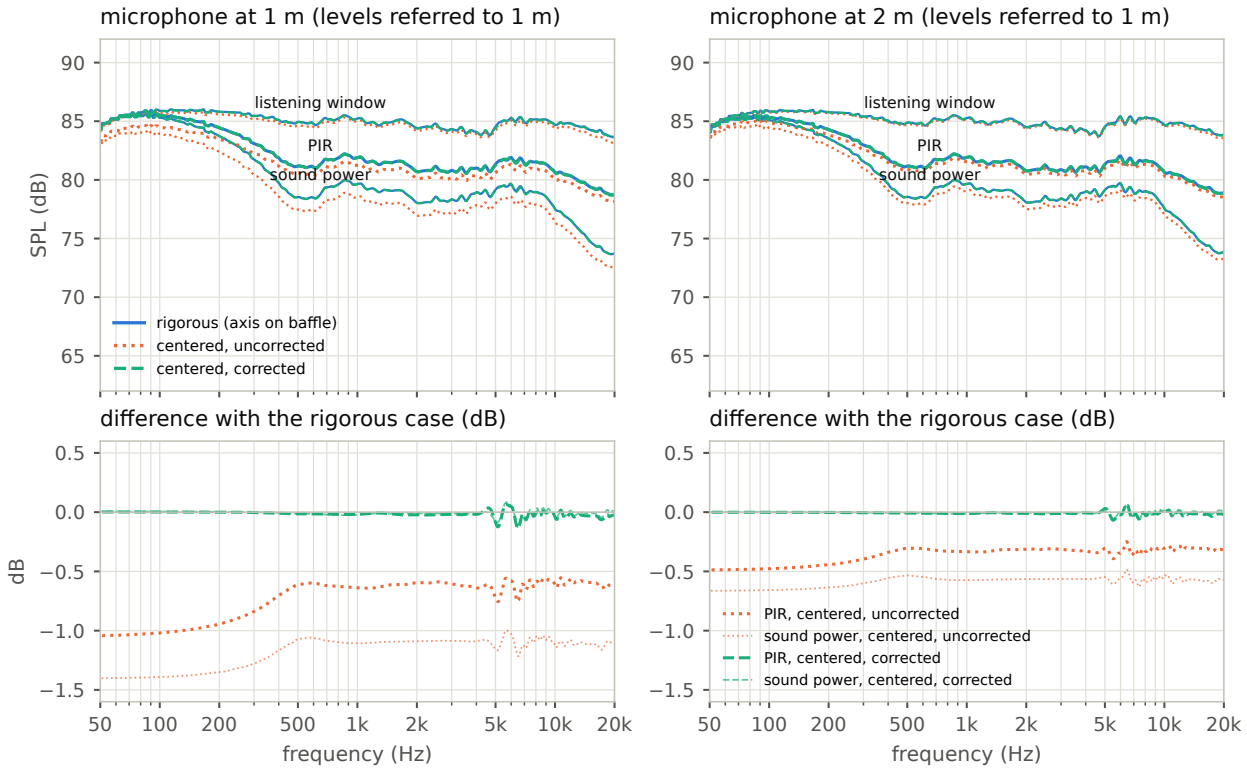


Figure 14: Top: listening window, PIR and sound power of the summed loudspeaker from the horizontal orbit (levels referred to 1 m). The corrected curves lie on the rigorous ones. Bottom: differences with the rigorous case.

the rotation axis, on the + side. Omitting it ($\ell = 0$, i.e. correcting only for the depth, as for a symmetric loudspeaker) leaves the summed loudspeaker 2.39 dB RMS off at 1 m and the PIR up to 3.82 dB off; entering it with the wrong sign is as bad (3.26 dB). For the PIR, and for the tweeter (4.13 and 5.16 dB RMS against 3.30 dB), both are worse than no correction at all (1.02 dB). The sign convention matters: ℓ is positive when the reference axis lies on the side the microphone faces at $+90°$. With the convention of this study (positive angles toward the tweeter side) the reference axis is on the + side; a turntable that counts angles the other way needs $\ell = -120$ mm and a woofer offset of +205 mm.

The woofer's own offset (−205 mm) matters less: omitting it raises the woofer's error from 0.09 to 0.27 dB RMS (1.7 dB at worst) at 1 m, and the summed loudspeaker hardly notices, the woofer being crossed over at 450 Hz where its directivity varies slowly.

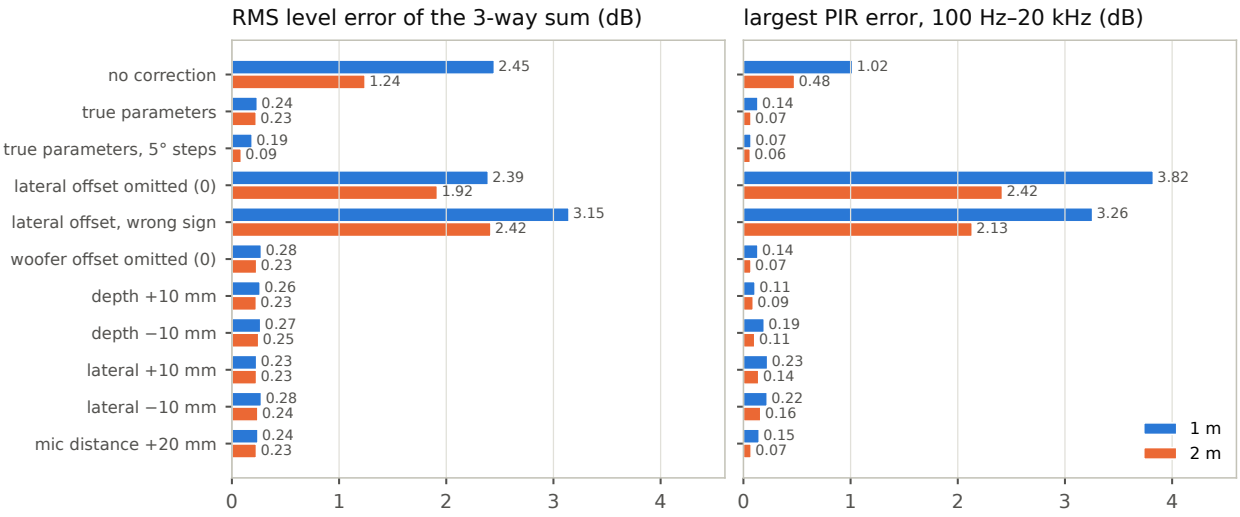


Figure 15: Effect of the correction parameters: RMS level error of the summed loudspeaker (left) and largest PIR deviation (right), at 1 m and 2 m.

Table 6: RMS level error per driver and for the summed loudspeaker, and largest PIR deviation, for the correction variants (dB).

	1 m					2 m				
	woofer	mid	tweeter	sum	PIR	woofer	mid	tweeter	sum	PIR
no correction	1.84	2.39	3.30	2.45	1.02	0.98	1.19	1.71	1.24	0.48
true parameters	0.09	0.21	0.40	0.24	0.14	0.03	0.14	0.41	0.23	0.07
true parameters, 5° steps	0.09	0.18	0.29	0.19	0.07	0.02	0.05	0.16	0.09	0.06
lateral offset omitted (0)	0.61	1.42	4.13	2.39	3.82	0.39	0.88	3.32	1.92	2.42
lateral offset, wrong sign	1.15	2.76	5.16	3.15	3.26	0.74	2.02	4.18	2.42	2.13
woofer offset omitted (0)	0.27	0.21	0.40	0.28	0.14	0.15	0.14	0.41	0.23	0.07
depth +10 mm	0.13	0.25	0.42	0.26	0.11	0.06	0.15	0.41	0.23	0.09
depth −10 mm	0.13	0.23	0.45	0.27	0.19	0.07	0.16	0.44	0.25	0.11
lateral +10 mm	0.12	0.18	0.40	0.23	0.23	0.05	0.14	0.42	0.23	0.14
lateral −10 mm	0.08	0.26	0.46	0.28	0.22	0.04	0.15	0.44	0.24	0.16
mic distance +20 mm	0.09	0.22	0.42	0.24	0.15	0.03	0.14	0.41	0.23	0.07

7 Discussion

The geometry is the easy part. For the direct sound of each driver the correction is exact up to interpolation, and the angular interpolation of a 10° orbit is good enough (below 0.1 dB RMS). The out-of-plane offsets between the drivers and the microphone, which the planar model ignores, do not matter at 1 m.

Diffraction is the limit, mostly behind the cabinet. The edge waves come from the baffle edges, not from the driver; the correction re-times them as if they did. In this simulation the consequence is small in front of the loudspeaker and grows behind it, at mid and high frequencies, where the edge waves are the response. Spatial averages are barely affected: the PIR of the corrected data is within about 0.1 dB of the rigorous one. Individual rear curves can show local deviations of a few dB in narrow bands.

1 m or 2 m. At 2 m the raw errors are halved and the finite-distance part of the diffraction error shrinks, but the angular-interpolation part does not; a finer orbit (5°) is then the effective remedy. In a real room the choice is also about the reflection-free time window, which shrinks with distance — outside the scope of this free-field study.

The lateral offset is the trap. The model handles the laterally shifted microphone, provided the lateral offset is entered, with the right sign. A wrong or missing lateral offset is worse than no correction. A quick check: at +90° the microphone must face the side counted as positive, and the reference axis of a layout like this one lies on that side if the microphone moves toward the tweeter at positive angles.

Limits of the study.

- Free field. The room, the gating window and its interaction with the correction (the delays change by up to 1 ms behind the cabinet) are not simulated.
- The DED model is an approximation: only the edges of the front baffle are modelled, the field behind the cabinet is the sum of the edge dipoles, and the edges are sharp. Real cabinets with rounded edges diffract less, which should help the correction.
- Rigid pistons and domes, acoustic centers on the baffle plane. A real acoustic center behind or in front of the baffle acts like a depth error for that driver; ±10 mm of depth error costs little here.
- Only the horizontal orbit. The vertical orbit, with the loudspeaker on its side on the turntable, follows the same geometry (with the driver offsets of that plane) and was not simulated; the CTA-2034-A curves above are horizontal-orbit estimates.

Checklist for a small-turntable polar measurement

1. Center the cabinet on the turntable; put the microphone on the reference axis (in front of the reference point) at 0°, at the reference height.
2. Measure the depth (rotation axis to baffle) and the lateral offset (rotation axis to reference axis) to within about 1 cm; note on which side the microphone faces at +90° and give the lateral offset that sign.
3. Give every driver its offset from the reference axis in the plane of rotation — a woofer beside the tweeter in a horizontal cabinet is not at 0.
4. Use 5° steps if the rear half matters at high frequency, particularly at 2 m.
5. Treat rear responses at mid and high frequencies as approximate (cabinet diffraction); spatial averages (sound power, PIR, DI) are robust.

8 Conclusion

On a small turntable the cabinet is centered and the microphone moved sideways to face the reference point; uncorrected, the off-axis data are too low (up to 2.5 dB behind the cabinet at 1 m), late and seen from the wrong direction, and the PIR is about 1 dB low. A geometric correction — each driver a point source with directivity, re-propagated and interpolated — restores them: essentially exactly for the direct sound, and to within a few tenths of a dB RMS once the cabinet's edge diffraction is included, the residual being concentrated behind the cabinet at mid and high frequencies. The spatial averages of CTA-2034-A, PIR included, come within about 0.1 dB of the rigorous ones. The method is only as good as its parameters, and one of them is critical: the lateral offset of the reference axis, with its sign.

A Additional figures

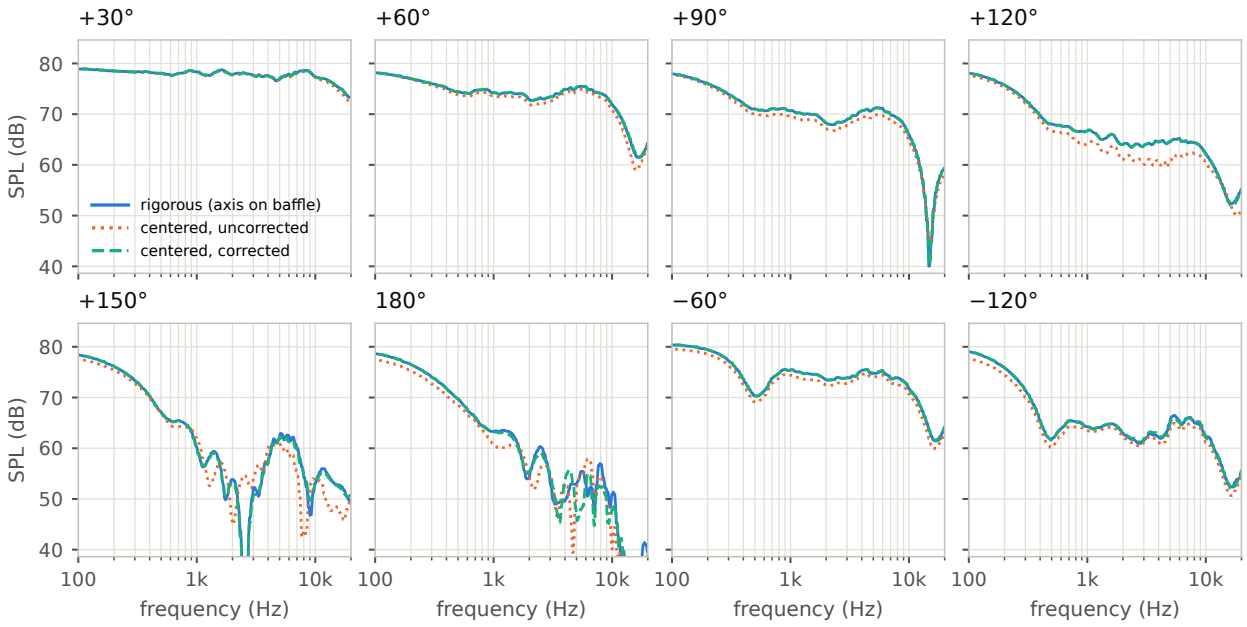


Figure 16: Summed loudspeaker at selected angles, 2 m (levels at 2 m), 1/12-octave smoothing.

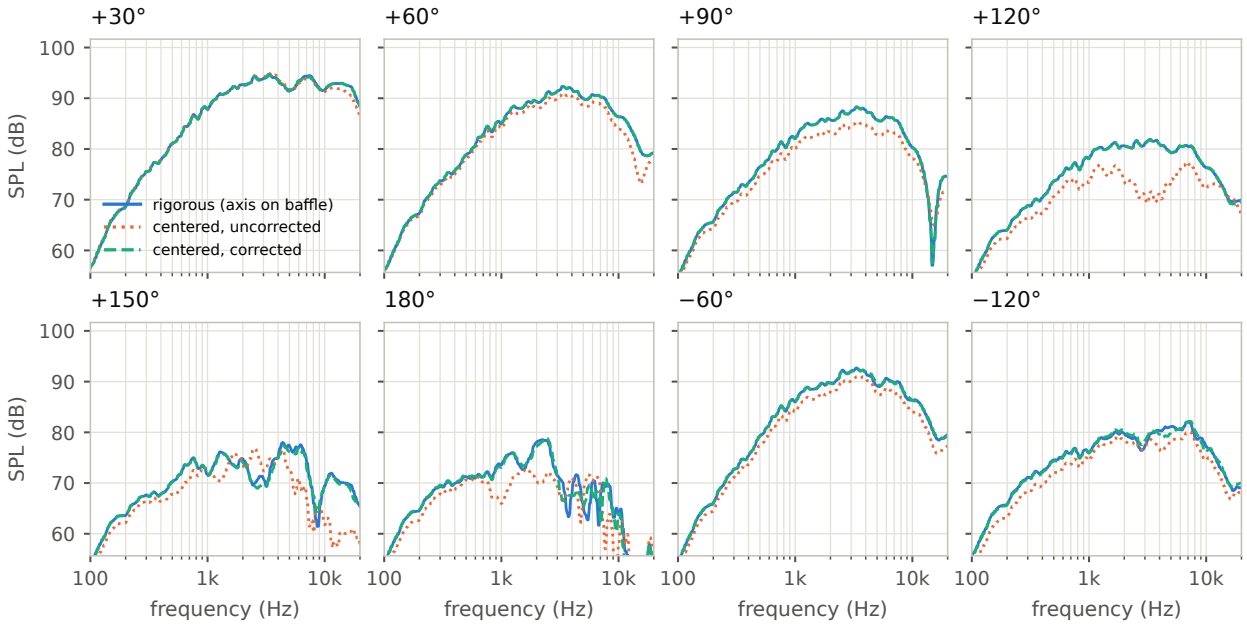


Figure 17: Tweeter alone at selected angles, 1 m, 1/12-octave smoothing. Behind the cabinet the response is made of edge waves; the corrected curves follow the rigorous ones with local deviations in the 4–8 kHz ripple.

B Reproducibility

Everything in this report is produced by the Python code of the repository (NumPy, SciPy, Matplotlib):

- `rotcenter/`: geometry and kinematics, radiators, driver models, DED diffraction, the correction, CTA-2034-A averages, crossover, metrics;
- `scripts/run_study.py`: all simulated measurements (≈ 5 min);
- `scripts/make_figures.py`, `scripts/make_schematics.py`: figures and tables;
- `tests/`: unit tests (kinematics, directivities, DED normalization, correction of an ideal point source).

The datasheet curves (`data/datasheets/`) were extracted from the vector graphics of the manufacturers' PDF datasheets; they belong to their manufacturers and are used here for analysis only.

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[1] M. Urban, C. Heil, C. Pignon, C. Combet, P. Bauman, "The Distributed Edge Dipole (DED) Model for Cabinet Diffraction Effects," *J. Audio Eng. Soc.*, vol. 52, no. 10, pp. 1043–1059, 2004.

[2] ANSI/CTA-2034-A, *Standard Method of Measurement for In-Home Loudspeakers*, Consumer Technology Association, 2015.

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